Tutorial-4 :-

$$\left(w_j^{new} = w_j^{old} + \eta \sum_{q=1}^{n} (y - \hat{y}_{pred})\, x_{qj} \right)$$

$$\left(z = w_0 + \sum_{q=1}^{n} w_q x_q \right)$$

$$\left\{ \text{if } \begin{array}{l} \hat{y}_{pred} \geq 0.5 \Rightarrow 1 \\ \hat{y}_{pred} < 0.5 \Rightarrow 0 \end{array} \right\}$$

$$\left(\hat{y}_{pred} = \sigma(z) = \frac{1}{1+e^{-z}} \right)$$

$$\left(w_0^{new} = w_0^{old} + \eta \sum_{q=1}^{n} (y - \hat{y}_{pred}) \right)$$

$$\left(\text{log loss} = -\frac{1}{n} \sum_{q=1}^{n} \left(y_q \log(\hat{y}_{pred}) + (1-y_q) \log(1-\hat{y}_{pred}) \right) \right) (\ln)$$

$j \rightarrow$

	W1 Suspicious words	W2 External links	W3 Email length	W4 Attachment	Spam
1	5	2	120	1	1
2	1	0	300	0	0
3	3	1	180	0	1
6	0	0	400	0	0
5	6	3	90	1	1

independent ⇒ suspicious words (x_1), external links (x_2), Email length (x_3) Attachment (x_4)

dependent ⇒ Spam (y)

1st attempt: $n=5$; $w_1 = w_2 = w_3 = w_4 = 0$; $w_0 = 0$

$$\Rightarrow z = 0 + \sum_{q=1}^{n} (w_q x_q) = 0$$

$$\hat{y}_{pred} = \frac{1}{1+e^{-0}} = \frac{1}{2} = 0.5 \Rightarrow 1$$

$$\log \text{loss} = \frac{-1}{5}\Big(1(\log(1)) + \cancel{(1-1)\log(1-1)}$$

$$+ 0\times\log(1) + (1-0)\log(1-1)$$

$$+ 1\times\log(0.5) + \cancel{(1-1)\log(1-1)}$$

$$+ 0\times\log(0.5) + (1-0)\log(1-1)$$

$$+ 1\times\log(0.5) + \cancel{(1-1)\log(1-1)}\Big)$$

$$\Rightarrow = \frac{-1}{\cancel{5}}\left(\left(\log 0.5\right)\times\cancel{5}\right) = -\log(0.5) = \log 2 = 0.693$$

$$(1/0.5)^{-1} \Rightarrow 2$$

2nd iteration

$$\Rightarrow w_1^{new} = 0 + (0.000001)\Big(\cancel{(1-1)5} + (0-1)1 + (1-1)3 + \cancel{(0-1)0}$$

$$+ \cancel{(1-1)6}\Big)$$

$$= \cancel{0 + (0.000001)(-1)} = -(10^{-6})\times 6.5$$

$$\Rightarrow w_2^{new} = 0 + (10^{-6})\Big(\cancel{(1-1)2} + \cancel{(0-1)0} + \cancel{(1-1)1} + \cancel{(0-1)0}$$

$$+ \cancel{(1-1)3}\Big)$$

$$= 3\times 10^{-6}$$

$$\Rightarrow w_3^{new} = 0 + (10^{-6})\Big(\cancel{(1-1)100} + \cancel{(0-1)300} + \cancel{(1-1)100} + \cancel{(0-1)400}$$

$$+ \cancel{(1-1)70}\Big)$$

$$= \cancel{7\times 10^{-6}}\ -155\times 10^{-6}$$

$$\Rightarrow w_4^{new} = 0 + (10^{-6})\Big(\cancel{(1-1)1} + \cancel{(0-1)0} + \cancel{(1-1)0} + \cancel{(0-1)0} + \cancel{(1-1)1}\Big)$$

$$= 10^{-6}$$

$$\Rightarrow w_0^{new} = 0 + (10^{-6})\left(\overset{0.5}{\cancel{(1-1)}} + \overset{-0.5}{\cancel{(0-1)}} + \overset{0.5}{\cancel{(1-1)}} + \overset{-0.5}{\cancel{(0-1)}} + \overset{0.5}{\cancel{(1-1)}}\right)$$

$$= \cancel{-2\times10^{-6}} \; \boxed{0.5\times10^{-6}}$$

(Data-1)

$$z = (0.5\times10^{-6}) + (-10^{-6}\times 66)5 + (3\times10^{-6})2 + (-155\times10^{-6})120 + (10^{-6})\times 1$$

$$= 10^{-6}\left(0.5 - \overset{-32}{\cancel{32.5}} + 5 - 18600 + 1\right)$$

$$= 10^{-6}\times(-18626)$$

$$= -0.018626$$

$$\hat{y}_{pred} = \frac{1}{1+e^{+0.018626}} = 0.49$$

$$\text{log loss} = \frac{-1}{5}\Big(1\log(0.49) + \cancel{(1-1)\log(1-0.49)}$$
$$+ \cancel{0\times\log(0.49)} + (1-0)\log(\overset{0.51}{\cancel{1-0.49}})$$
$$+ 1.\log(0.49) + \cancel{(1-1)\log(1-0.49)}$$
$$\cancel{+ 0.\log(0.49)} + (1-0)\log(\overset{0.51}{\cancel{1-0.49}})$$
$$+ 1.\log(0.49) + \cancel{(1-1)\log(1-0.49)}\Big)$$

$$= \cancel{\frac{-1}{5}\left(\log(1-0.49)\times 5\right)}$$

$$= \cancel{0.693}\; 0.693$$

3rd Iteration

$$w_1^{new} = (-6.5\times10^{-6}) + 10^{-6}\big((1-0.49)5 + (0-0.49)1 + (1-0.49)3 + (0-0.49)0 + (1-0.49)6\big)$$

$$= 10^{-6}(0.15)$$

$$w_2^{new} = (3\times10^{-6}) + 10^{-6}\big((1-0.49)2 + \cancel{(0-0.49)0} + (1-0.49)1 + \cancel{(0-0.49)0} + (1-0.49)3\big)$$

$$= 3\times10^{-6} + 10^{-6}(3.06)$$

$$= 6.06\times10^{-6}$$

$$w_3^{new} = (-155\times10^{-6}) + 10^{-6}\big((1-0.49)120 + (0-0.49)300 + (1-0.49)180 + (0-0.49)400 + (1-0.49)90\big)$$

$$= (-155\times10^{-6}) + 10^{-6}\big((0.51)390 + (0.49)(700)\big)$$

$$= -299.1(10^{-6}).$$

$$w_4^{new} = 10^{-6} + 10^{-6}\big((1-0.49)1 + \cancel{(0-0.49)0} + \cancel{(1-0.49)0} + \cancel{(0-0.49)0} + (1-0.49)1\big)$$

$$= 10^{-6} + 10^{-6}(0.51\times2)$$

$$= 2.02\times10^{-6}$$

$$w_0^{new} = 0.5\times10^{-6} + 10^{-6}\big((1-0.49) + (0-0.49) + (1-0.49) + (0-0.49) + (1-0.49)\big)$$

$$= 1.05\times10^{-6}$$

$$z = w_0 + w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4$$

$$= 1.05\times10^{-6} + (0.15\times10^{-6})1 + (6.06\times10^{-6})0 + (-299.1\times10^{-6})300$$

$$+ \cancel{((2.02)\times10^{-6})\times0}$$

$$= 10^{-6}(1.05 + 0.15 - 89730)$$

$$z = -89{,}728.8\times10^{-6}$$

$$\hat{y}_{pred} = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{89{,}728.8\times10^{-6}}} = 0.477$$

$$\text{log loss} = \frac{-1}{5}\Big(1.\log(0.47) + \cancel{(1-1)\log(1-0.47)}$$

$$\cancel{+0.\log(0.47)} + (1-0)\log(1-0.47)$$

$$+1.\log(0.47) + \cancel{(1-1)\log(0.47)}$$

$$\cancel{+0.\log(0.47)} + (1-0)\log(1-0.47)$$

$$+1.\log(0.47) + \cancel{(1-1)\log(1-0.47)}\Big)\Big)$$

$$= \frac{-1}{5}\left(3\ln(0.47) + 2\ln(0.53)\right)$$

$$= 0.706$$